

MODELING HOLOCENE CHANGES IN THE LOCATION AND ABUNDANCE OF BEECH POPULATIONS IN EASTERN NORTH AMERICA

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(Received October 8, 1986)

Abstract

Dexter, F., Banks, H.T. and Webb III, T., 1987. Modeling Holocene changes in the location and abundance of beech populations in eastern North America. *Rev. Palaeobot. Palynol.*, 50: 273–292.

Maps of Holocene changes in pollen percentages show changes in the magnitude and location of taxon abundances. They also show changes in the patterns of decreasing abundance around the areas of maximum abundance. Transport equations with source/sink, advection, and diffusion terms provide analogs for modeling these different components of continent-scale population dynamics. We used recently developed numerical techniques to fit such a transport equation to Holocene changes in beech (*Fagus*) pollen percentages along a transect between Tennessee and southern Quebec. A constant coefficient model explained 91% of the variance for the changes in beech pollen percentages between 8000 and 500 yr B.P. Estimated values for the coefficients were 9.6 km²/year for diffusion, 0.05 km/yr for advection, and 0.18/1000 yr for the exponent in the source/sink term. Perturbation and residual analysis showed these values to be robust and statistically sound. A second version of the transport model with temporally varying parameters for the source/sink term yielded small but significant improvement by explaining 94% of the variance. A plausible interpretation of the model results is that the changes in beech populations reflect spatial shifts in the multifactorial environmental conditions that are most favorable for the growth of beech trees.

Introduction

Pollen data from spatial networks of radio-carbon-dated sediment cores record late-Quaternary continent-scale movements of plant populations in time and space. Maps of these distributional changes show the range limits of many taxa expanding across the landscape, the abundance maxima moving from region to region, and the actual abundances of individual taxa increasing or de-

creasing through time as the plants flourish or perish in different regions. This division of the movement and changes into components has analogs in models used in several disciplines, including transport models in the physical sciences and biodiffusion models in the biological sciences (Okubo, 1980; Levin, 1981; Karieva, 1982; Banks and Karieva, 1983). We have used recently developed numerical techniques (Banks and Lamm, 1985) to fit a transport model to pollen data that record the abundance changes in beech (*Fagus*) populations within eastern North America. In this initial application we have used data from a transect of palynological sites along the main axis of

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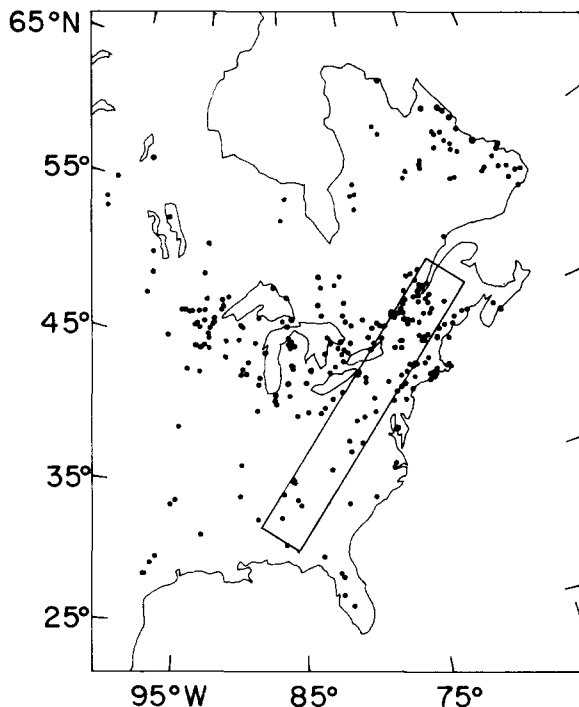


Fig.1. Map of eastern North America showing the location of sites with pollen data and the location of the transect from Alabama to southern Quebec along which data were chosen for plotting Fig.3.

Holocene beech-population movement between the southeastern U.S.A. and the St. Lawrence River Valley (Fig.1).

Since Agassiz's introduction of the idea of continental glaciation in 1837, many scientists have described the movements of plant populations during ice advance and retreat (Darwin, 1859). Skellam (1951) was the first to apply a transport model with a diffusion term to the postglacial range extension of plant populations. For his calculations, he used rough estimates for the time interval and distance over which oak populations expanded their range in England. Since then, these estimates have been revised, and such estimates can now be based on dense networks of pollen diagrams with radiocarbon dates, which are available for the British Isles (Birks et al., 1975; Bennett, 1985), Europe (Huntley and Birks, 1983), the Soviet Union (Peterson, 1983), and eastern North America (Davis, 1976, 1981; Bernabo and

Webb, 1977; Bartlein and Webb, 1985; Delcourt and Delcourt, 1981). The maps of pollen percentages illustrate the changing rates and directions of the postglacial spatial dynamics of tree taxa and include range-boundary extensions as a special case (Webb et al., 1983a, b; Gaudreau and Webb, 1985).

The postglacial movement of beech populations has attracted much attention and discussion since Davis (1976) originally used isochrone maps to illustrate "migration" rates for several tree taxa. Since then, several studies have included maps for beech (Bernabo and Webb, 1977; Kapp, 1977; Davis, 1978, 1981, 1983; Davis et al., 1986; Bailey and Ahearn, 1981; Webb et al., 1983a, b; Bennett, 1985; Gaudreau, 1986). These studies have illustrated the northward movement of beech populations into the northeastern United States and their somewhat later westward extension into the Midwest. In his recent review, Bennett (1985) also illustrated an increase in beech populations within the Southeast between 12,000 and 10,000 yr B.P.

The mapped percentages for beech pollen illustrate three main components of motion: (1) movement of the abundance maximum; (2) a spreading out of values from the maximum; and (3) changes in the maximum value. Within the disciplines of hydrology, meteorology, and oceanography, models with advection, diffusion, and source/sink terms are commonly used to represent such multicomponent patterns of change. These models are derived from a mass-balance equation by including a source/sink term that approximates the way that the "mass" of the modeled entity is added or subtracted to the system (Banks and Palatt, 1975). The advection and diffusion terms model the movement and spreading of the entity between locations in the study area, and the source/sink term allows for increases and decreases in the amount of the entity at individual locations. For data derived from plant taxa, the source/sink term can record either the net result of birth and death at each location or (for trees) the net addition or subtraction of biomass, volume, or basal area.

The movement of tree populations across the landscape results from as yet unspecified biogeographical processes, for which the terms in the transport model may provide a reasonable analog in the same way that molecular diffusion models provide analogs for describing turbulent diffusion in the atmospheric boundary layer. If a simple model involving diffusion, advection, and source/sink terms can fit the data, then this model may help suggest mechanisms involved in the biogeographical movement. The relative importance among the terms in the simple model may also provide insights. We present a first test of these possibilities.

Data and data analysis

We chose a data set that posed an interesting but relatively simple problem in order to explore the possibility of fitting a transport model to pollen data. We first consulted maps that show the changes in beech pollen percentages (Fig.2). From these, we selected a transect

of sites aligned in the geographic direction that shows the maximum changes in beech populations (Figs.1 and 3). We thus reduced the dimensions of the data set from 3 (2 spatial and 1 temporal) to 2, but retained a data set that included significant population changes. We next restricted the spatial and temporal range of the data along the transect in order to obtain a subset that represented a single population expansion with smooth and mainly monotonic changes (Figs.4 and 5).

The data used to produce Figs.1–3 came from a set of over 300 pollen diagrams available in computer files at Brown University. Bartlein and Webb (1985) provide a list of the sites with data at 6000 yr B.P., which includes sites used to produce Fig.1. Webb et al. (1983a, b) and Gaudreau and Webb (1985) describe the methods used (1) to obtain interpolated pollen percentages at each 1000-year interval and (2) to produce the isopoll maps (Fig.2). The pollen percentages used in our study were based on a sum of all tree, shrub, and herb pollen.

The transect along the main axis of move-

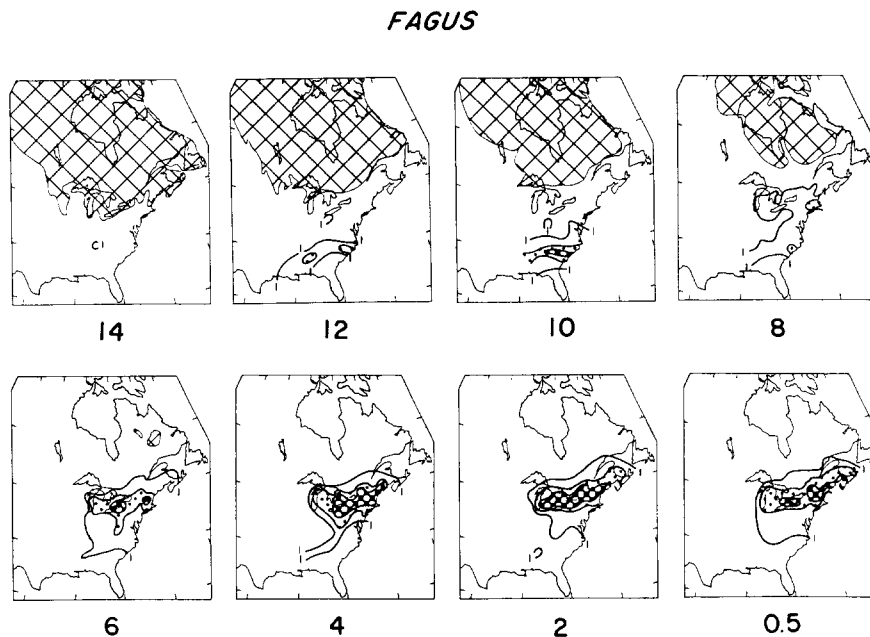


Fig.2. Maps with isofrequency contours (isopolls) for 1.5, 10% *Fagus* (beech) pollen from 14,000 to 500 yr B.P. Broad cross-hatching marks area of Laurentide ice sheet.

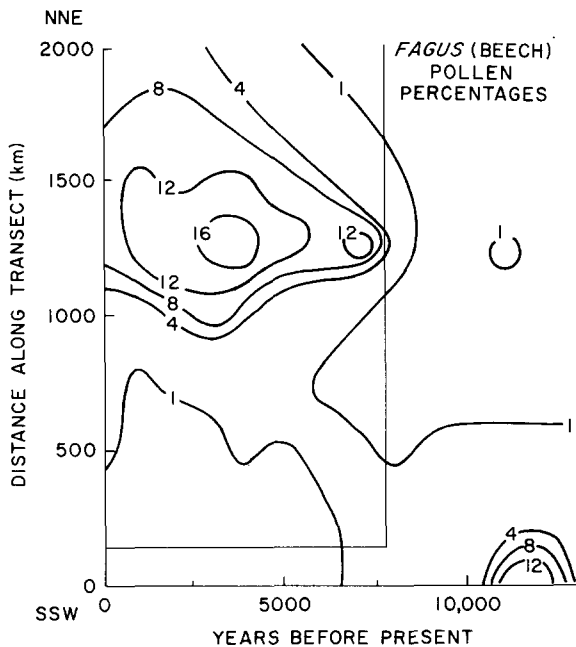


Fig. 3. Time-space diagram of the smoothed percentages for *Fagus* (beech) pollen from the sites along the transect in Fig. 1.

ment for beech populations included all sites within a 200-km wide transect from Alabama to Quebec (Table I; Fig. 1). We then eliminated 3 sites (Big Pond, Criders Pond, and Panther Run Pond) whose location in areas of high

relief made them under-represent the regional amount of beech pollen around their location along the transect. We also eliminated 5 other sites that were located so close to adjacent sites that their inclusion might have caused numerical difficulties in fitting the model. In order to avoid major human disturbance to the record, we excluded all data younger than 500 yr B.P. Since that date, logging and agriculture have altered the vegetation and decreased the abundance of beech pollen (Gaudreau and Webb, 1985).

The full transect of the data from 13,000 to 500 yr B.P. shows the initial increase in beech pollen percentages in the Southeast (lower right corner, Fig. 3). The increase in percentages in the Northeast occurred after 8000 yr B.P. and was most pronounced in New York and New England, about two-thirds of the way north along the transect (Fig. 3). This northern expansion of the beech populations provided pattern enough for testing the model. We therefore restricted the data set to the upper left-hand portion of the time-space plot (Figs. 3-5). This data set shows that beech abundances increased both northwards and southwards and that the location of the highest abundances moved north (Fig. 5).

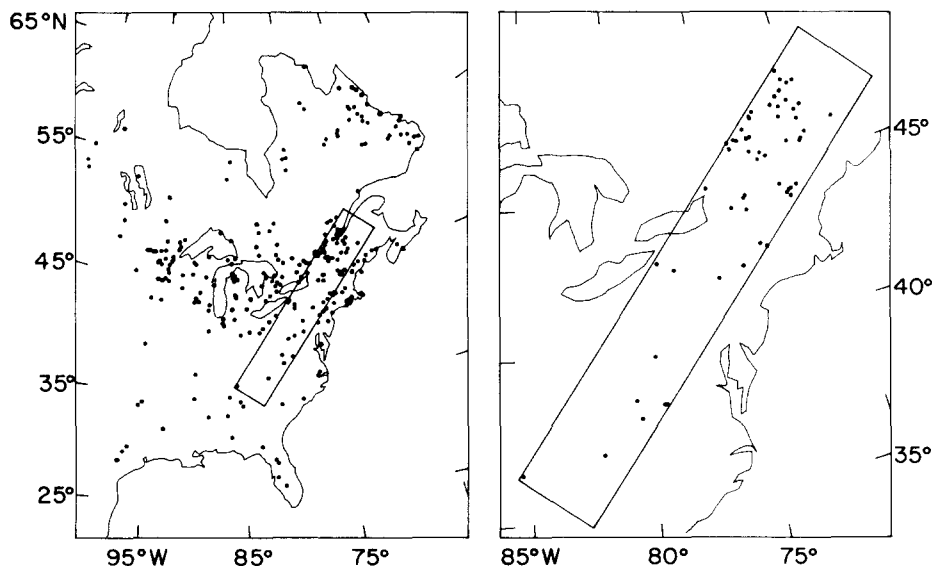


Fig. 4. Map showing the transect of sites with pollen data used in estimating the parameters in the model and in constructing Fig. 5.

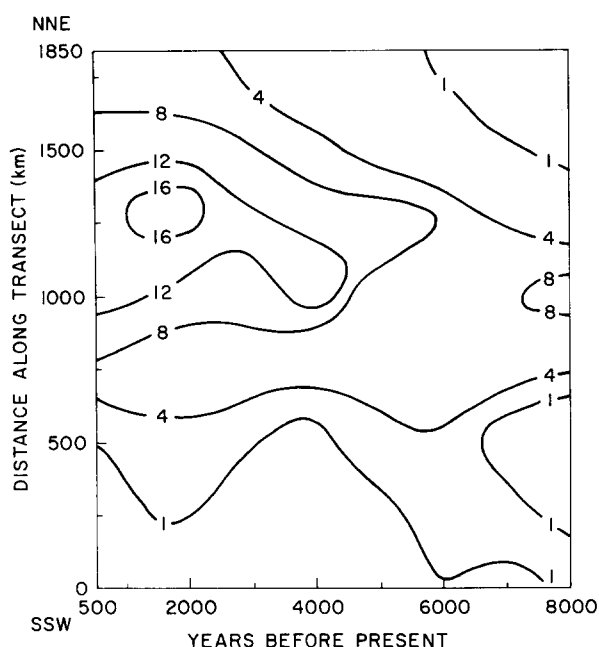


Fig.5. Time-space diagram of the percentages of *Fagus* (beech) pollen from Tennessee (0 km) to southern Quebec (1850 km). See Fig.4.

Pollen percentages

In our calculations with the transport model, we used beech pollen percentages as proxies for the abundance of beech trees along the transect. Previous work has shown that pollen percentages are approximately linearly related to the percentages of trees within the forests about each pollen site (Webb et al., 1981; Bradshaw and Webb, 1985; Prentice and Webb, 1986). When the forest cover remains relatively uniform near each site, the tree percentages can serve as a measure of absolute as well as relative abundance and are best related to such abundance measures as canopy area, basal area, or growing stock volume. The relatively low amount of herb pollen at all sites along the transect from 8000 to 500 yr B.P. indicates that the forest cover was relatively uniform around each site. Progress is being made toward obtaining the data and methods for converting pollen percentages into quantitative measures of tree abundance, but this work is still too

experimental for such conversion factors to be used in our study.

Contours

The contours on the time-space plots (Figs.3 and 5) reflect two steps of analysis. The data were first divided into grids of 9 (space) by 14 (time) for Fig.3 and of 8 by 8 for Fig.5. Within each grid cell, the median pollen percentage was calculated, and contours were drawn by bivariate linear interpolation between median values. The median values were also used as data for the program with which we estimated the parameters and coefficients in the transport model. Because no sites were located in the fourth spatial cell from the south end of the transect (Fig.4), data were only available for 56 grid points, and the 7 values for the cells with the oldest date were used as initial values in the model (Fig.6).

Model and estimation procedures

We used a model of the form

$$\begin{aligned} \frac{\partial \rho(t, x)}{\partial t} + \frac{\partial}{\partial x} \left[V(t, x) \rho(t, x) \right] \\ = \frac{\partial}{\partial x} \left[D(t, x) \frac{\partial \rho(t, x)}{\partial x} \right] + \alpha(t, x) \rho(t, x) \end{aligned} \quad (1)$$

in which the rate of change of the beech abundances $\rho(t, x)$ through time is related to the sum of an advection term, a diffusion term, and a source/sink term, where t and x represent temporal and spatial coordinates, $V(t, x)$ is an advection coefficient, $D(t, x)$ is a diffusion coefficient, and $\alpha(t, x)$ is the coefficient for the source/sink term. Within this model, the temporal changes in beech abundances are assumed to result (a) from transport processes that are either diffusion-like over an area or involve advection in a certain direction, or (b) from in situ increases or decreases depending on whether either the birth rate exceeds the death rate or unaccounted for sink/source processes are active. These latter processes could include, for example, movement into the transect from the side.

TABLE I

Sites with pollen data along the transect

Site name	Latitude	Longitude	State or province	Elevation (m)	Relative distance from end of transect (a)	Relative distance from end of transect (b)
Cahaba Pond	33 34 N	86 31 W	AL	204	0.000	
Quicksand	34 19 N	84 52 W	GA	285	0.076	
Mingo Pond	35 9 N	86 12 W	TN	300	0.079	
Pigeon Marsh	34 40 N	85 10 W	GA	630	0.083	
Anderson Pond	36 2 N	85 30 W	TN	305	0.135	0.000
Shady Valley Peat	36 31 N	81 56 W	TN	383	0.249	0.131
Potts Mountain Pond	37 36 N	80 8 W	VA	840	0.342	0.239
Cranberry Glades	38 12 N	80 17 W	WV	1029	0.363	0.264
Quarles Pond	37 59 N	79 4 W	VA	500	0.387	0.291
Hack Pond	37 59 N	79 0 W	VA	451	0.389	0.293
Buckle's Bog	39 34 N	79 16 W	MD	814	0.448	0.362
Big Pond	39 46 N	78 33 W	PA	634	0.475	
Crider's Pond	39 58 N	77 33 W	PA	290	0.510	
Panther Run Pond	40 48 N	77 25 W	PA	634	0.548	
Rose Lake	41 55 N	77 55 W	PA	631	0.582	
Houghton Bog	42 32 N	78 40 W	NY	428	0.589	0.524
Belmont Bog	42 15 N	77 55 W	NY	497	0.596	0.532
Protection Bog	42 37 N	78 28 W	NY	430	0.598	
Ely Lake	41 46 N	75 50 W	PA	385	0.630	0.572
Balsam Lake	42 1 N	74 36 W	NY	762	0.673	0.621
Heart's Content Bog	42 14 N	73 58 W	NY	107	0.698	
West Sand Lake Peat Bog	42 38 N	73 36 W	NY	170	0.724	0.681
Berry Pond	42 30 N	73 19 W	MA	600	0.726	0.684
Brandreth Lake	43 55 N	74 41 W	NY	583	0.747	0.708
Lambs Pond	44 39 N	75 48 W	ON	107	0.749	0.710
Clear Pond	43 45 N	74 1 W	NY	510	0.758	0.720
Upper Wallface Pond	44 9 N	74 3 W	NY	945	0.773	0.738
Heart Lake	44 11 N	73 58 W	NY	664	0.777	0.742
Deer Lake Bog	44 2 N	71 50 W	NH	1325	0.827	0.799
Mirror Lake	43 57 N	71 42 W	NH	200	0.827	0.800
Bugbee Bog	44 22 N	72 9 W	VT	398	0.831	0.805
Lac Tania	45 46 N	74 18 W	PQ	305	0.832	0.806
Kinsman Pond	44 8 N	71 44 W	NH	1140	0.833	0.807
Eagle Lake Bog	44 10 N	71 40 W	NH	1275	0.836	0.811
St. Germain	45 56 N	74 22 W	PQ	473	0.838	0.812
Farnham Bog	45 17 N	72 59 W	PQ	55	0.846	0.822
Lac des Atocas	45 32 N	73 18 W	PQ	120	0.848	
Lac a la Tortue	45 32 N	73 19 W	PQ	137	0.848	
Lac Bouleaux	45 33 N	73 19 W	PQ	126	0.848	0.825
Gea-I — St. Hippolyte	45 59 N	73 59 W	PQ	365	0.849	0.825
Lake of the Clouds	44 16 N	71 19 W	NH	1542	0.850	0.826
St. Calixte	45 57 N	73 52 W	PQ	261	0.851	0.827
Yamaska	45 28 N	72 52 W	PQ	265	0.856	0.834
Mont Shefford	45 21 N	72 35 W	PQ	282	0.859	0.837
Lac Romer	45 58 N	73 20 W	PQ	20	0.865	0.843
Lanoraie, St. Henri Bog	45 59 N	73 18 W	PQ	18	0.866	0.845
Lanoraie, St. Joseph Bog	45 59 N	73 18 W	PQ	18	0.866	0.845
Lanoraie, St. Jean Bog	46 0 N	73 13 W	PQ	20	0.869	0.848
Gabriel	46 16 N	73 28 W	PQ	250	0.873	0.853

TABLE I (continued)

Site name	Latitude	Longitude	State or province	Elevation (m)	Relative distance from end of transect (a)	Relative distance from end of transect (b)
Lac Castor	46 36 N	72 59 W	PQ	220	0.899	0.883
Sam	46 39 N	72 58 W	PQ	240	0.901	0.885
Albion	45 40 N	71 19 W	PQ	320	0.904	0.889
Mauricie	46 47 N	72 50 W	PQ	270	0.910	0.895
Boundary	45 34 N	70 41 W	ME	603	0.917	0.904
Unknown Lake	45 37 N	70 38 W	ME	489	0.920	0.908
Dosquet	46 27 N	71 30 W	PQ	140	0.930	0.919
Dufresne	45 51 N	70 21 W	PQ	650	0.937	0.927
St. Raymond	46 53 N	71 48 W	PQ	160	0.939	0.930
St. Benjamin	46 17 N	70 36 W	PQ	330	0.947	0.939
Base de Plein Air de St. Foy	46 47 N	71 20 W	PQ	16	0.948	0.940
Marcotte	47 4 N	71 25 W	PQ	503	0.957	0.950
Petit Lac Terrien	46 35 N	70 37 W	PQ	404	0.958	0.952
St. Jean, Ile d'Orleans	46 56 N	70 56 W	PQ	68	0.964	0.958
Joncas Lake	47 15 N	71 10 W	PQ	780	0.970	0.965
Joncas Bog	47 16 N	71 10 W	PQ	747	0.970	0.966
Lac Colin	46 43 N	70 18 W	PQ	658	0.972	0.967
Upper South Branch Pond	46 5 N	68 54 W	ME	300	0.984	0.981
Malbaie	47 36 N	70 58 W	PQ	800	0.989	0.987
Baie St. Paul — Ange	47 28 N	70 41 W	PQ	640	0.991	0.990
Montagnais	47 54 N	71 10 W	PQ	800	0.995	0.995
Lac Mimi	47 30 N	70 22 W	PQ	411	1.000	1.000

(a) Full data set (Fig.1).

(b) Data used in analysis (Figs.4, 5).

The derivation of eq.1 from mass balance considerations is outlined in the Appendix. The formal assumption of mass balance in the transport model links spatial changes within a region to temporal changes at a point. Diffusion and advection are just two components of the spatial dynamics (see Appendix); the addition of the source/sink term in eq.1 allows for a lack of mass balance. We note that α is unlikely to be equal to zero because plant populations do not in fact conserve mass. Spatial and temporal change in plant populations, therefore, involves no direct causal connection, and we must caution against too literal an interpretation of the material transport paradigm. A more appropriate interpretation of the mechanisms represented in the model is in terms of changes in climatic or environmental conditions that either promote or inhibit growth of the taxon being modeled.

These conditions may be controlled by climatic changes that are diffusive on a regional scale and advective on a regional to continental scale.

For physical realism, the diffusion coefficient in our model should always be positive, but the advection coefficient can be positive or negative, depending upon the direction of advective movement and the convention used to represent a given direction. We let positive $V(t, x)$ imply movement northwards along the transect.

Simulation example

We performed a simulation exercise in order to illustrate what each term in eq.1 might contribute to the general pattern of beech pollen percentages in Fig.5. Simulations were run with four different versions of the model, and each simulation began with its initial

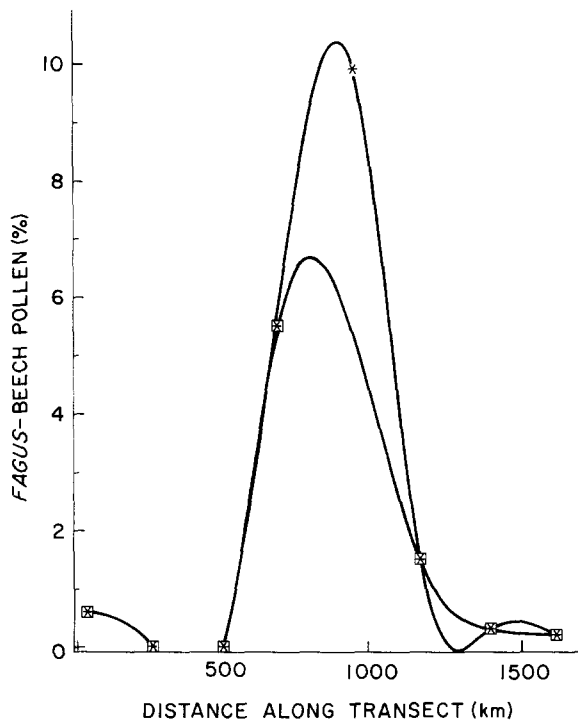


Fig.6. Initial conditions used in fitting the model to the data. Curves fit to *Fagus* (beech) pollen percentages vs. distance for 8000 yr B.P. The original data are marked with a star, and the data points used to test the robustness of the estimated parameters are in squares. To alter the initial conditions, the largest observed was deleted.

conditions specified by the data (Fig.6). For the simulation, the values of the coefficients D , V , and α were constant in time and space. The first version contained only the diffusion term and was of the following form:

$$\frac{\partial \rho(t, x)}{\partial t} = D \frac{\partial^2 \rho(t, x)}{\partial x^2} \quad (2)$$

The second version contained both the diffusion and advection terms, the third version had diffusion and sink/source terms, and the fourth version included all three terms. Each version was run with values of the constant coefficients that we estimated (in a manner explained below) from the data (Table II). The simulations provide concrete examples that are an excellent illustration of the use of the model and the role of each of its terms.

In the first simulation, the abundances

spread out symmetrically, and the highest values decreased through time (Fig.7.A). The addition of the advection term (its positive coefficient indicates northward movement) led to a northward drift in the symmetric pattern of the first simulation (Fig.7.B). The third version (with both diffusion and source/sink terms) switched the pattern from opening to the right to opening to the left because a positive source/sink term allowed the total "mass" of beech trees to increase through time (Fig.7.C). The final simulation with all three terms included the northward drift of the area of highest abundances (Fig.7.D) and provided a good representation of the pattern in Fig.5.

Estimation procedures

We considered eq.1 for $0 < x < 1850$, $500 < t < 8000$, with boundary conditions $\rho(t, 0) = \rho(t, 1850) = 0$ (which are compatible with our data) and initial conditions $\rho(0, x) = \Phi(x)$ where Φ is assumed known (given in Fig.6 for the investigations under discussion here). For this system we assumed a data set $\{\hat{\rho}_{ij}\}$ of observations of beech densities ρ at points (t_i, x_j) . We let $q = (D, V, \alpha)$ denote the unknown coefficients in eq.1 which we sought to estimate from the observations of beech pollen percentages. This leads to an inverse problem or parameter estimation problem based on a least squares fit-to-data criterion: choose from some admissible parameter set Q a value q so as to minimize

$$J(q) = \sum_{i,j} |\rho(t_i, x_j, q) - \hat{\rho}_{ij}|^2 \quad (3)$$

where $\rho(t, x, q)$ is the value at (t, x) of the solution of eq.1 corresponding to q .

In general, this problem should be considered in cases where the parameter set Q consists of a set of functions D , V , α that can vary with space, time, or both. Even in the situation where these coefficients are assumed to be constants, this problem cannot be solved analytically. Indeed such problems can be difficult computationally and require care in the choice of methods for obtaining solutions

TABLE II

Constant coefficient model: estimates and tests of the solutions

Analysis	Coefficients	Search range	Initial guess	Estimate	Variance explained (%)
Original analysis	$D(\text{km}^2/\text{yr})$	0.0–228	114	9.6	91
	$V(\text{km}/\text{yr})$	0.0–0.5	0.25	0.05	
	$\alpha(10^{-3}\text{yr}^{-1})$	0.0–0.8	0.53	0.18	
New initial guess	D	0.0–228	0.46	8.7	91
	V	0.0–0.5	0.002	0.05	
	α	0.0–0.8	0.13	0.17	
New search range	D	0.0–228	114	9.1	91
	V	–0.5–0.5	0.25	0.05	
	α	–0.8–0.8	0.53	0.18	
New search range and initial guess	D	0.0–46	0.46	8.7	91
	V	0.0–0.25	0.002	0.05	
	α	0.0–0.4	0.13	0.17	
Altered initial conditions	D	0.0–228	114	7.3	90
	V	0.0–0.5	0.25	0.05	
	α	0.0–0.8	0.53	0.22	
Gaussian noise added	D	0.0–228	114	8.7	90
	V	0.0–0.5	0.25	0.04	
	α	0.0–0.8	0.53	0.18	

on the computer. Banks and Kareiva (1983), Banks and Lamm (1985), and Banks et al. (1985) describe the approach we have used. It entails an iterative technique that produces a sequence of estimates converging to a minimizer for eq.3. We used a version of the Levenberg-Marquardt scheme that is a modified Gauss-Newton algorithm. Our procedure also involves some type of method for numerically solving the partial differential eq.1. In generating the results reported in this manuscript, we used a Galerkin state approximation scheme based on cubic B-splines along with a version of Gear's variable order/variable step size routine to solve the resulting approximating ordinary differential equations. Full technical details and explanations are given in the above cited references.

As we have already noted, Q should in general contain nonconstant functions. Because an iterative search cannot be carried out over a space of functions on a digital

computer, we therefore had to approximate (or to parameterize in some finite dimensional manner) the coefficient functions D , V , α . In our calculations with the beech data, we represented the coefficient functions in terms of linear combinations (anywhere from 2 to 6 terms) of linear spline elements (in time or space) and carried out searches for the best parameters in the expansions for the coefficients. For example, a spatially varying advection coefficient might be represented as $V(x) = \sum_j v_j l_j(x)$, where the l_j are linear spline elements, and scalar parameters v_j are searched for in order to obtain a "best" velocity. In our results, therefore, the parameters that we sought to estimate were either one of the coefficients D , V , α (in the case in which we only considered constant function coefficients) or a set of scalar parameters that determine the approximate representation for the variable coefficient being estimated. We also considered fits-to-data in which one of the

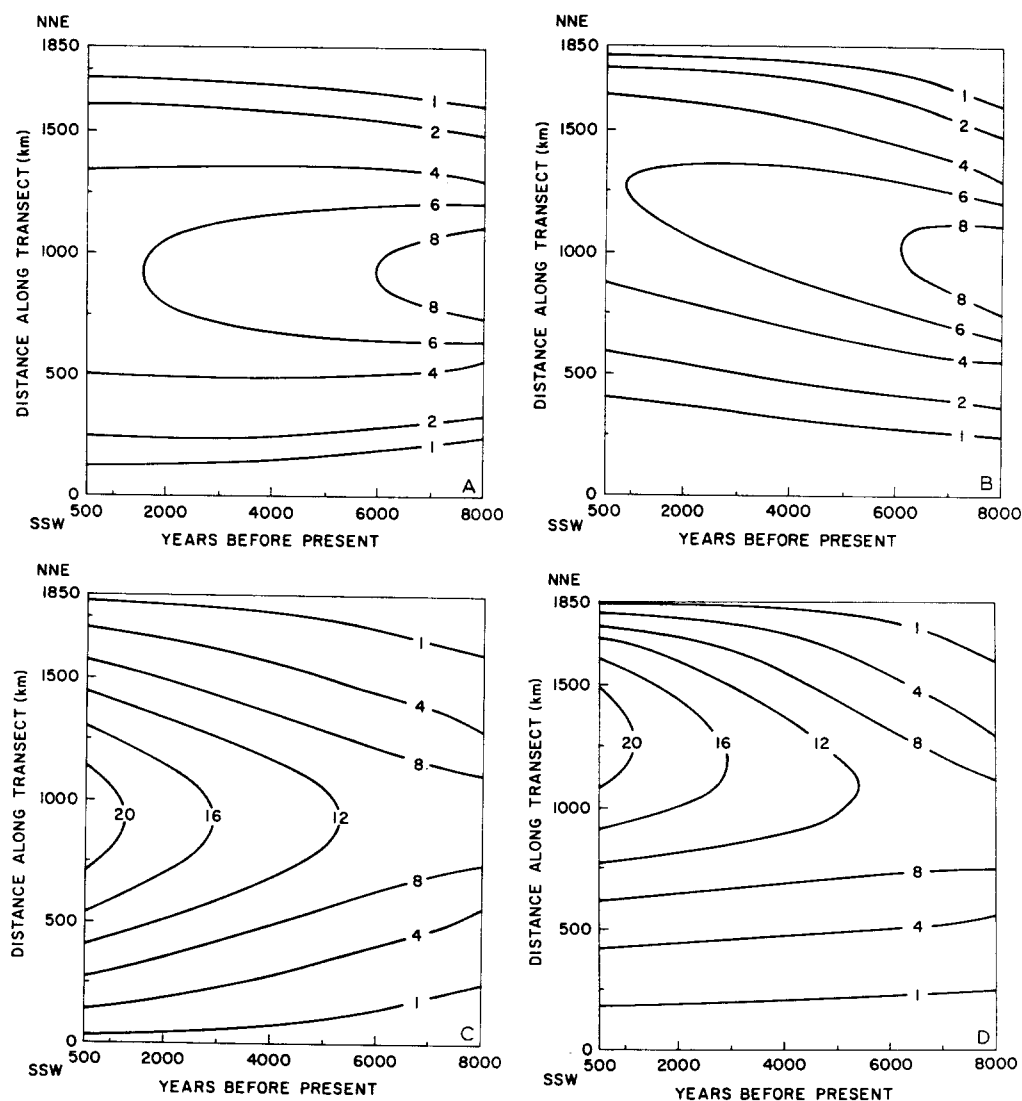


Fig. 7. Set of simulated results using the estimated values for the constant coefficients (Table II) (a) when only the diffusion coefficient is non-zero, (b) when both diffusion and advection coefficients are non-zero, (c) when both diffusion and source/sink coefficients are non-zero, and (d) when all three coefficients are non-zero.

coefficients was assumed constant while another varied in time or space (Table III).

The mathematical and computational soundness of the above procedures requires considerable study, and a rigorous mathematical theory of convergence and stability of the methods that we used can be developed. Such findings are a special case of theoretical results given in Banks and Lamm (1985), Banks et al. (1985), and Banks (1985).

Tests for robustness

Several procedures were adopted to test for robustness of the parameter estimates. These included changing the initial guesses and search ranges (in the iterative search algorithm) for the parameters, perturbing the initial conditions, and adding Gaussian noise to the data (Table II). The estimated parameters were considered robust if their values remained

TABLE III

Model solutions and tests with and without the convection term

Solution	Coefficient	Search range	Initial guess	Estimate ^a	Variance explained (%)
1	$D(\text{km}^2/\text{yr})$	0.0–230	114	13.2	57
	$V(\text{km}/\text{yr})$			0.0	
	$\alpha(10^{-3}\text{yr}^{-1})$	–0.8–0.8	0.53	0.18	
2	D			13.2	59
	V			0.0	
	$\alpha(t)$	–0.8–0.8	0.18	(–0.17, 0.29, 0.42 0.00, 0.34, –0.46)	
3	D			13.2	88
	V			0.0	
	$\alpha(x)$	–2.7–2.7	0.18	(–0.29, 0.55)	
4	D			13.2	90
	V			0.0	
	$\alpha(x)$	–2.7–2.7	0.18	(–1.55, 0.20, 0.46)	
5	D			13.2	90
	V			0.0	
	$\alpha(x)$	–2.7–2.7	0.18	(–0.97, –0.20, 0.37 0.24)	
6	D			13.2	90
	V	–1.0–1.0	0.05	0.005	
	$\alpha(x)$			(–1.53, 0.20, 0.46)	
7	D			13.2	92
	V	–1.0–1.0	0.01	0.05	
	$\alpha(x)$	–2.7–2.7	0.18	(0.45, 0.26, 0.06)	
8 ^b	D			13.2	94
	V	–1.0–1.0	0.01	0.11	
	$\alpha(x)$	–2.7–2.7	0.18	(0.10, 0.47, –0.37)	
9 ^c	D			13.2	91
	V	–1.0–1.0	0.01	0.06	
	$\alpha(x)$	–2.7–2.7	0.18	(–0.44, 0.32, –0.06)	

^aEstimates are for constant coefficients or for the parameters of the variable coefficients.^bAltered initial conditions.^cGaussian noise added.

essentially unchanged after each of these alterations to the data. Standard tests for the residuals of the model were also applied.

Other tests of the results involved modifications to the model by the addition or subtraction of terms to the equation and by variation of the parameterization of each of the coefficients. In judging the different models, we, of course, prefer those with the fewest parameters that explain the most variance. We also desire that

the model have robust coefficient (or parameter) values that are also biologically and physically reasonable. We used an F-test to help us judge whether the addition of parameters significantly decreased the residual sum of squares (RSS) and thus significantly increased the corresponding explained variance. We recognized that our data did not fulfill the assumptions required by the F-test, but we followed the practice of Banks and Kariëva (1983) in using

the F-test in an ad hoc manner to measure the approximate significance of the decrease in RSS. We rejected any models that yielded negative diffusion coefficients because such are physically and conceptually unrealistic. We also rejected solutions in which selected parameters exceeded certain empirically set limits for diffusion or advection.

Results

We began with tests of the model to see how the value of α affected the choices of D and V and vice versa. For a variety of choices of D , V , and α , we used our data (Fig.5) to calculate the RSS and, if the RSS was smaller than 2200 (180% of the total sum of squares), we recorded the values for D , V , and α . These tests showed that as α increased, so also did D or V or both D and V (Fig.8). The increased values for D and V allowed the model to cope with the increased rate of addition of trees implied by higher values of α . The test showed the non-uniqueness of any particular solution to the model and required us to put appropriate limits on the biological reasonableness of D and V . As a guide, we used the empirical estimate for beech range-extension of 0.2 km/yr given in Davis (1981). We allowed for the possibility of short-term periods of higher rates by setting upper limits of 230 km²/yr for D and 0.5 km/yr for V (Table II).

Constant coefficient solution

We first fitted a model with constant coefficients using the upper limits to set the search range on D and V . We also used Fig.8 to set limits for α . A solution of $D=9.6$ km²/yr, $V=0.05$ km/yr, and $\alpha=0.18/1000$ yr explained 91% of the variance and proved robust when subjected to the sequence of tests listed in Table II. The initial conditions were altered by removing the largest value (Fig.6), and Gaussian noise (mean of zero, variance of 0.25%) was added to the value at each grid cell. None of these changes lead to significant changes in the parameters or explained variance (Table II). The residuals for the model proved to be

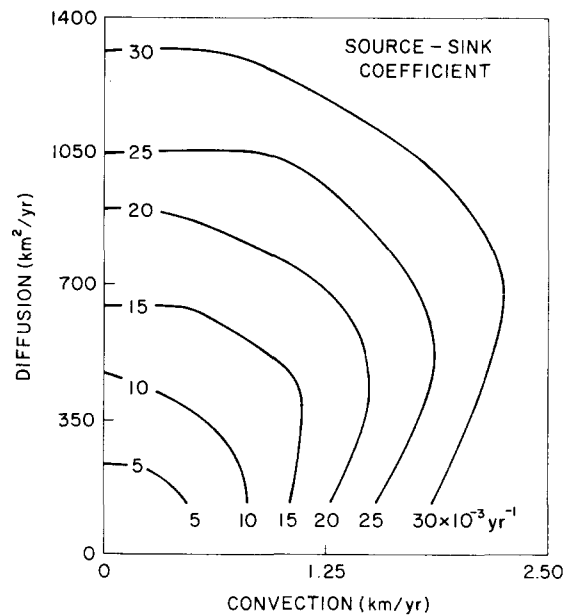


Fig.8. Plot showing the relationship among the values for diffusion, and source/sink. Least squares fit of simulated constant coefficient model was less than 2200.

normally distributed with a mean of zero, and the residuals were uncorrelated with the model predictions and with the spatial and temporal coordinates.

The constant-coefficient model reproduced the pattern of beech pollen percentages in Fig.5 quite well (Fig.7.D). Each of the three terms accounts for a different aspect of the pattern. The values for D and V are much lower than either of their upper limits (Table II) or the value of 0.2 km/yr estimated by Davis (1981). This fact indicates that the coefficients are "tuned" to the high values for beech and reflects the relative insensitivity of the model to the changes in the low-valued contours that monitor the position of the range boundary. (Note that finding a fit to the high values explains much more variance than is explained by a good fit to the low values.) Davis' (1981) estimate of 0.2 km/yr was for the rate at which the range boundary moved. The model result indicates that the average rate of diffusion for the beech abundances was about 10 km²/yr and that the population center moved northward

over the 7500-year interval at a rate of about 50 m/yr. This latter rate is in close agreement with the one that can be calculated directly from Fig.5 by noting that the region of highest abundances for beech pollen shifted northward about 300 km in 7500 years.

Possible models with variable coefficients

Some patterns evident in the plot of the data (Fig.5) do not appear in Fig.7.D. For instance, the two temporal decreases in the area with highest abundance are not evident in Fig.7.D. The first occurs between 8000 and 7000 yr B.P. and the second occurs after 1000 yr B.P. (Fig.5). This observation led us to two sets of additional model runs. The first set involved a selection of runs to see if a model with spatially or temporally varying values of α (source/sink) might render the advection term unimportant. The second set of runs tested a series of models with spatially or temporally varying values of each coefficient to see if a new model could be found that improves on the constant-coefficient model with its explained variance of 91% (Table II).

(1) Test of advection term

The series of model tests to see whether the advection term was necessary began with the version of the constant-coefficient model with just diffusion and source/sink terms (Table III). Without advection, both D and α had slightly different values from those in the model in which advection was included, but the new model explained only 57% of the variance and was significantly inferior to the model that included advection (Table II; Fig.7.D). A model with diffusion and a temporally variable source/sink coefficient was equally inferior (Table III), but the model with a constant diffusion term and a spatially variable source/sink term explained almost the same amount of variance as the original 3-term model (Table III). Three versions of this model were investigated, one with 2 parameters in the linear representation for $\alpha(x)$, a second with 3 parameters, and a third with 4 parameters.

Because the third produced no improvements over the second (Table III), the second model was used for further testing. When advection was allowed to be non-zero in the second model with its parameters preset by the previous run, this model offered no improvements, and the coefficient for the advection was almost zero. A second run of this model in which the parameters for V and $\alpha(x)$ were both allowed to vary yielded a value of V similar to that in the original model (Table II) and a slight improvement in the explained variance (Table III). The coefficient estimates obtained in this model varied, however, when the initial conditions were modified or Gaussian noise was added to the data (Table III). These results suggest that the original constant coefficient model is preferable to the four-parameter D and $\alpha(x)$ model and that the advection term is an important component of the model when it is applied to the data in Fig.5.

(2) Variable coefficient models

Our final tests with the data were to see whether a variable coefficient model could do significantly better than the constant coefficient model. We ran six tests in which each coefficient was allowed to vary with either time or space while the other coefficients were preset at their values in the constant coefficient model (Table IV). Each of these tests involved an 8-parameter equation with single parameters for each constant coefficient and six parameters for each coefficient that varied in time or space. Only the model with a time-variable source/sink term proved to be an acceptable improvement over the constant coefficient model (Table IV). This model explained just enough more variance to be significant (94 vs. 91%), its parameter values were biologically reasonable, and its parameter values were insensitive both to changes in the initial conditions and to the addition of Gaussian noise to the data. All other variable-coefficient models failed one or more of these tests, and the model with spatially variable diffusion gave the worst results (Table IV).

The parameter values for the temporally

variable source/sink term $[\alpha(t)]$ vary from negative at the beginning and end to positive in the middle (Fig.9). The negative coefficients at 8000 and 500 yr B.P. allow for the slight decrease in abundance that the data show at these times (Fig.5). Because the new model can reproduce these subtle features in the data, it explains a small but significant amount more of the total variance.

Discussion

Our objective was to discover what could be learned by fitting a transport model with diffusion, advection, and source/sink terms to the spatio-temporal patterns of pollen percentage changes during the past 18,000 years (Figs.2 and 3). Because maps of pollen data from radiocarbon-dated sediment cores illustrate broad-scale patterns of change in taxon abundances, we studied the maps of several pollen types for a taxon with a relatively simple pattern of continuous change during this time period. The pattern of beech pollen changes during the past 8000 years (Fig.5) met this criterion and provided a good first example for testing both the model and the computational procedures for fitting the model to pollen data (Fig.7; Tables II–IV). The pollen data indicated (a) that the area with abundant beech trees increased in size and (b) that within this area the number of beech trees also increased (Fig.5).

A constant-coefficient version of the transport model was able to fit this pattern quite well (Fig.7.D, Table II), and the expansion of the model to allow for temporal changes in the source/sink coefficient yielded an even better fit to the data (Table IV). This expanded version of the model was able to reproduce some minor decreases in abundance at about 8000 and 500 yr B.P. No other variable-coefficient model yielded robust, biologically plausible results that significantly improved on the constant-coefficient model. Our results also showed that the constant-coefficient model produced better results than a model without

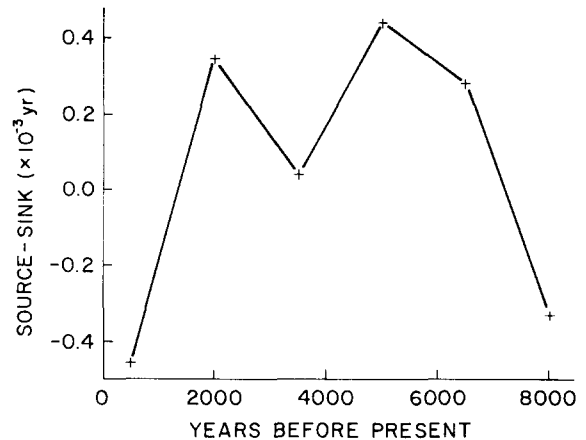


Fig.9. Plot of estimated parameters when the source/sink term varies with time (see Table IV).

advection but with a spatially varying source/sink coefficient.

The relative success of the constant-coefficient model raises the question of whether its diffusion, advection, and source/sink terms correspond to distinct vegetation processes. The maps and time-space diagrams (Figs.2, 3 and 5) illustrate a complex pattern of change that can be broken down conceptually into three main components. These are: (1) a symmetrical spreading outwards of the region with high abundances (Figs.5 and 7.A), (2) a drift northwards in the area of high abundances (Fig.7.B, D) and (3) local and regional increases and decreases in abundance (Figs.2, 5 and 7.C, D). Can these three components be usefully described as representing diffusion, advection, and the net effects of abundance increases and decreases? The answer depends upon what level of abstraction is appropriate in interpreting the maps of the pollen data. Three levels are possible. Mapped pollen data could conceivably represent (1) the production and dispersal of pollen, (2) the increase (or decrease) in plant population size and the changes in location of the range boundary and the population center, or (3) the changes in the favorability and location of the environment suitable for the establishment and growth of beech trees.

At the first level of abstraction, the maps for

TABLE IV

Tests of variable coefficient models^a

Diffusion (km ² /yr)	Convection (km/yr)	Source sink (10 ⁻³ yr ⁻¹)	Variance explained (%)	F-test	Initial conditions	Gaussian noise	Biologically reasonable
$D(t)$	0.05	0.18	95	+	-	+	-
$D(x)$	0.05	0.18	92	-	-	-	-
9.58	$V(t)$	0.18	92	-	-	+	+
9.58	$V(x)$	0.18	93	-	-	-	+
9.58	0.05	$\alpha(t)$	94	+	+	+	+
9.58	0.05	$\alpha(x)$	92	-	+	+	+

^a(Each coefficient was allowed to vary in either time or space and had 6 parameters.) Plus signs (+) indicate significance, non-sensitivity or reasonableness. Minus signs (-) indicate the opposite.

beech pollen (Fig.2) might be viewed as representing the 14,000-year record of the production, diffusion, and advection of a continuous cloud of pollen. While in the air, pollen grains are subjected to advective and diffusive motion; and on a daily and regional scale, pollen production, transport, and accumulation involve diffusive, advective, and/or source/sink processes (Gregory, 1973; Solomon and Harrington, 1979). These time and space scales are *not* represented on the maps (Fig.2), however, because the mapped data reflect a long-term integration of the annual production and transport of pollen grains.

The second level of abstraction for pollen data is that of plant populations. Here pollen abundances are used as a proxy for plant numbers (Prentice, 1985), and seeds must be imagined as the physically mobile entities (Good, 1931). Most plants after all are sessile organisms. But the isopoll maps show that the regions with high plant abundances "move" and vary in size through time (Figs.2, 3 and 5). How do they do it? What aspect of the plants or populations is being transported from Georgia to Quebec as the 5% contour for beech pollen shifts northward? Seeds may move, but their movement is not a sufficient explanation because pollen abundances reflect plant abundances, not seed density. We know that certain aspects of seed transport are diffusive (Johnson and Adkisson, 1985), and we can even

imagine that the migration of particular bird species can provide an advective mechanism. For example, if the birds fly north when the seeds are available, but return south when they are unavailable, the net result could be a northward transport of seeds. But the advective transport of seeds is *not* required to explain a directional shift in the area of high abundances, once a population is established. The seed source exists, and the abundance change reflects a change in the success of the plants in one taxon relative to the plants of other taxa.

Just as pollen settles from the air at rates too fast for the maps to record, so too do the seeds in general move at rates much faster than the mapped abundance changes. Pollen and seed transport occurs, but neither is directly recorded by the 2000-year time steps in the maps (Fig.2). What the maps illustrate can be visualized as the movement of the environment favoring plant growth for each taxon. This interpretation of the data involves yet another level of abstraction for pollen data. At this level, the changing abundance in pollen data is seen as a proxy for the changing patterns in the environmental conditions most conducive to the growth of the plants that produce the pollen. Changes in these conditions across a region alter the competitive balance among the taxa in the forests and lead to abundance and compositional changes.

This "growth" environment is a combination of seasonal temperature and precipitation conditions, and its dynamics depend on the joint and often correlated behavior of all the variables involved. It will move in a direction (i.e. advectively) when the combination of variables moves in that direction, and it will spread out symmetrically in response to the random variations in storm track position or air mass frequency that are required to produce the right combination of temperature and precipitation conditions.

Factors other than environmental change are sometimes invoked to explain long-term vegetational change (see discussion in Ritchie, 1986; Prentice, 1986). For example, some hypotheses about late Quaternary vegetation change weight heavily the arrival of a new taxon whose presence then alters the competitive balance in the forests and affects the realized niche for each of the other taxa present (Birks, 1981; Davis, 1981). The pollen evidence for beech indicates that beech trees were present for centuries and even millenia before major increases in their population numbers occurred (Fig.2; Bennett, 1985; Gaudreau, 1986). This record suggests that the arrival of beech trees was not a sufficient condition to cause major increases in their abundance. Soil development can be another factor affecting beech population change, but the broad scale of the changing patterns in beech pollen indicates that soil development is not a major factor, because the region with high abundances of beech trees includes a wide mixture of soil types. Changes in fire frequency are yet another factor to consider, but temperature and moisture conditions control long-term fire frequency, especially at the spatial scale shown on the maps. This discussion of factors that control competitive relationships among taxa points to climatic change as the most likely cause of the long-term continental-scale changes in pollen abundance that are shown on the isopoll maps (Fig.2).

The growth environment for beech trees is an entity that can diffuse and be advected. It can also change in favorability and thus

induce source/sink-type changes. Advection of the growth environment may occur as the continent is heated radiatively and as warm, moist air masses extend further northward for longer periods. The growth environment can spread out symmetrically and change in favorability as frontal conditions develop and storms begin tracking along the front. Variations in storm frequency, position, and intensity will affect precipitation conditions and make the growth environment more or less favorable for beech growth over a broad area. At this level of abstraction, the transport equation is modeling the behavior of the growth environment; and diffusion, advection, and source/sink changes are real processes affecting its location and extent.

A key assumption in discussing the pollen record at the third level of abstraction is the claim that the vegetation is in dynamic equilibrium with climate. Webb (1986) has argued for this assumption, and the evidence that Bennett (1985) and Gaudreau (1986) present for changes in beech populations suggest that this assumption is valid, certainly since 8000 yr B.P. in the Northeast. Concern with disequilibrium conditions has centered on how seed dispersal might limit range extension (Davis et al., 1986). The main focus of our results, however, has been on the movement of the population center where seeds are readily available, thus rendering seed dispersal unimportant as a mechanism for creating disequilibrium conditions.

The interpretation of pollen data in terms of vegetational and environmental change has long been an area for critical analysis and debate (Birks, 1981; Prentice, 1983, 1986; Ritchie, 1986). Our interpretation is that continent-scale pollen maps reflect vegetation changes that are approximately in dynamic equilibrium with climate and result from spatial shifts (advection and diffusion) in the growth environment of individual taxa.

From this perspective, we infer that the environment favorable for beech trees in the Northeast appeared by 8000 yr B.P., expanded its areal coverage, while shifting its center slowly northward after 7000 yr B.P., and then

became slightly less favorable after 500 yr B.P. (Figs.3 and 5). For the westward movement of beech into the Midwest after 10,000 yr B.P. (Fig.2), the growth environment for beech appears to have been advected westward and then to have spread out symmetrically north and south while favoring an increase in the amount of beech trees and pollen.

Dynamic behavior in microfossils compared

Movements of tree populations can be compared usefully with those of planktonic and benthic foraminifera. Both of these types of organisms show abundance and location changes during the Quaternary that are similar in magnitude and speed to those in either the maps or the time-space diagrams of pollen data (e.g. Figs.2 and 3; Ruddiman and McIntyre, 1976; Ledbetter and Balsam, 1985). Unlike the pollen records, however, in which the fossil entities (pollen grains) are mobile and different from the living sessile entities (the plants), the fossil record for planktonic foraminifera preserves the calcified parts of the living organisms that are mobile while alive. Intermediate between these two data sets is the record of benthic forams in which the fossil tests are directly derived from the living, but sessile, organisms. Relative to planktonic foraminifera, maps of pollen data therefore provide a less direct record of the living organisms that produced them. The mobility of the living organisms makes it easier to imagine how populations of planktonic forams move from place to place and thus track the environment that is favorable to their growth. Because neither benthic forams nor plants can themselves move, it is harder to visualize how their populations move hundreds of km. The fact that they do move these distances on a time scale similar to that for planktic foraminifera implies that neither the sessile nature of the organisms nor the indirect reading of the living organisms (by pollen) are important factors influencing the nature of the dynamic processes that are recorded.

The dynamic behavior of areas where a

planktic species is highly abundant can serve as a model for understanding the dynamics of similar areas for tree populations. While alive, the foraminifera are susceptible to advection and diffusion in the water masses in which they grow. The mobility of living forams means that their populations can be moved hundreds of km in one year. The evidence, however, suggests that the areas with high numbers of individuals remain relatively fixed in location for hundreds of years and only move slowly over thousands of years. Each year the "mass" of each species of foraminifera stays relatively fixed while water flows through it. On a transect through the main population center, one will observe relatively high birth rates in its center and but relatively high death rates with low birth and growth rates at the boundaries. The regions of high density for each species, therefore, mark areas of the ocean in which the environmental conditions are most suitable for that species' growth. When over thousands of years this growth environment moves, so does the area of maximum growth for each species of foraminifera. The time-space plot of foraminifera will record the movement of this growth environment through time (Ruddiman and McIntyre, 1976). By analogy, we can visualize the long-term record of pollen data as providing a similar record of tree populations and their growth environments.

Time-space diagrams.

The model was used to fit the data displayed in time and space. For the initial tests, we restricted the data to one spatial dimension, but more advanced versions of the calculation procedure will allow application of the model to the data in three dimensions (two spatial and one temporal). Recent work by Stead (1983), Stead and Webb (in prep.), and Bartlein et al. (in prep.) allows the data to be interpolated in three dimensions and will aid this future modeling effort. The inverse-problem techniques that we have used are, in principle, applicable to the higher dimensional problems. However, our initial efforts in this direction suggest that some care must be taken with the computational

aspects as well as higher dimensional model development. For example, computational instabilities tend to increase with model dimension. Also, higher dimensional models can accommodate good fits with different parameter values, sometimes making it difficult to ascertain which of the suggested mechanisms might be responsible for certain movement patterns.

The two-dimensional plots of the data are informative in their own right and deserve further use. McAndrews (1966), Cushing (1967), and Wright and Watts (1969) all constructed time-space transect plots of pollen zones or inferred vegetation regions in Minnesota. Correctly oriented transects provide key summaries for many critical patterns of change that maps may obscure. Plots like Figs.3 and 5, which contain contours for pollen percentages, illustrate patterns critical to the understanding of past vegetational dynamics.

Our work to prepare time-space diagrams for beech showed that different levels of smoothing the current data set can affect the patterns evident in the data. For example, a 9×9 grid of the data from Table I and Fig.4 produced a data set with maximum values at two separate locations in the northeast after 4000 yr B.P. Such a data set would yield somewhat different estimates for the coefficients in the transport model. This fact means that the selection of the data set and the choice of the interpolation and smoothing methods will affect the results obtained from applying the transport model. Palynologists should therefore concern themselves with understanding and testing the choices that control the representation of the data in the time-space diagrams. These choices include the location and density of sites with pollen data, the dating control at each site, the inclusion of data from sites in anomalous locations (e.g. at high elevations), and the interpolation and smoothing methods used. Our methods and analysis were adequate for this initial application and test of the transport model, but future applications will profit from improved procedures for the selection and analysis of the data. (See: Note added in proof, p. 292)

Acknowledgments

This research was supported by the National Science Foundation through grants from the Climate Dynamics Program (ATM84-06832) and the Mathematics Program (MCS-8205355). It was also supported by the Air Force Office of Scientific Research under Contract No. AFOSR-84-0398. We thank R. Arigo, J. Avizinis, T. Judd, K. Murphy, L. Zia, and M. Prestas for technical assistance and P.J. Bartlein, G.L. Jacobson, Jr., and I.C. Prentice for critical readings of early drafts of this paper. We thank numerous palynologists who gave T. Webb permission to use their data. D.C. Gaudreau, P. Richard, R. Spear and D.R. Whitehead kindly supplied unpublished pollen counts.

Appendix

Equation 1 in the text is derived from the equation of mass continuity (or mass balance), which equates temporal changes to spatial variations and is written as

$$\frac{\partial \rho(t, x)}{\partial t} + \frac{\partial [\rho(t, x) V(t, x)]}{\partial x} = 0 \quad (\text{A1})$$

When applied to the atmosphere, this equation states that density $\rho(x, t)$ or mass per unit volume, which varies in time (t) and space (x), is conserved. For our data set, the "mass per unit volume" being conserved is some measure of taxon abundance per unit area, e.g. tree density, biomass, or basal area. (We referred to $\rho(t, x)$ in the text as "tree abundance.") In eq.A1, $\partial/\partial t$ and $\partial/\partial x$ are partial derivatives and $V(t, x)$ is the velocity in the x direction. Equation A1 indicates that tree abundance changes at any point must result from spatial variations in either the tree abundance (ρ) or velocity (V) fields. A source/sink term $[r(t, x)]$ is added to the equation to allow for the addition and/or loss of total tree abundance within the transect. An equation of this form must be written for each taxon i , and we thus obtain a system of equations.

$$\frac{\partial \rho_i(t, x)}{\partial t} + \frac{\partial [\rho_i(t, x) V_i(t, x)]}{\partial x} = r_i(t, x) \quad (\text{A2})$$

We may then distinguish two components of the taxon mass flux

$$\rho_i V_i = \rho_i(V_i - V) + \rho_i V \quad (\text{A3})$$

where V is the advective velocity of the environment in which the plants grow. Equation A3 divides $\rho_i V_i$ into a diffusive-type term $[\rho_i(V_i - V)]$ and an advective-type term $(\rho_i V)$. By assuming Fick's first law of diffusion for the diffusion term, i.e., $\rho_i(V_i - V) = -D_i \partial \rho_i / \partial x$ and that $r_i = \alpha_i \rho_i$, i.e., the addition or loss in tree density is Malthusian in form, we obtain the following equation:

$$\begin{aligned} \frac{\partial \rho_i(t, x)}{\partial t} + \frac{\partial}{\partial x} [V(t, x) \rho_i(t, x)] \\ = \frac{\partial}{\partial x} \left[D_i(t, x) \frac{\partial \rho_i(t, x)}{\partial x} \right] + \alpha_i(t, x) \rho_i(t, x) \end{aligned} \quad (\text{A4})$$

In this paper, we consider only one taxon, and therefore omit the subscripts in (A4). The final form of the transport equation depends heavily on tacit assumptions (Fick's law, Malthusian growth) that we have made concerning the mathematical form that best represents the diffusion-like and source/sink mechanisms.

References

- Bailey, R.E. and Ahearn, P.J., 1981. A late- and postglacial pollen record from Chippewa Bog, Lapeer Co., MI: further examination of white pine and beech immigration into the central Great Lakes region. In: R.C. Romans (Editor), *Geobotany II*. Plenum Press, New York, N.Y., pp.53-74.
- Banks, H.T., 1985. On a variational approach to some parameter estimation problems. LCDS Tech. Rep. 85-24, Brown Univ., Providence, RI.
- Banks, H.T. and Karieva, P., 1983. Parameter estimation techniques for transport equations with applications to populations dispersal and tissue bulk flow models. *J. Math. Biol.*, 17: 253-273.
- Banks, H.T. and Lamm, P.D., 1985. Estimation of variable coefficients in parabolic distributed systems. *IEEE Trans. Autom. Control*, AC-30: 386-398.
- Banks, H.T. and Palett, P.J., 1975. *Mathematical Modeling in the Biological Sciences*. LCDS Lecture Notes 75-1, Brown Univ., Providence, RI.
- Banks, H.T., Karieva, P.M. and Lamm, P.D., 1985. Modeling insect dispersal and estimating parameters when mark-release techniques may cause initial disturbances. *J. Math. Biol.*, 22: 259-277.
- Bartlein, P.J. and Webb III, T., 1985. Mean July temperature at 6000 yr B.P. in eastern North America: regression equations for estimates from fossil pollen data. *Syllogeus*, 55: 301-342.
- Bartlein, P.J., Webb III, T. and Prentice, I.C., in prep. New climatic response surfaces for eastern North American pollen data.
- Bennett, K.D., 1985. The spread of *Fagus grandifolia* across eastern North America during the last 18,000 years. *J. Biogeogr.*, 12: 147-164.
- Bernabo, J.C. and Webb III, T., 1977. Changing patterns in the Holocene pollen record of northeastern North America: a mapped summary. *Quat. Res.*, 8: 64-96.
- Birks, H.J.B., 1981. The use of pollen analysis in the reconstruction of past climates: a review. In: T.M.L. Wigley, M.J. Ingram, and G. Farmer (Editors), *Climate and History*. Cambridge Univ. Press, Cambridge, pp.111-138.
- Birks, H.J.B., Deacon, J. and Peglar, S., 1975. Pollen maps for the British Isles 5000 years ago. *Proc. R. Soc. London B*, 189: 87-105.
- Bradshaw, R.H.W. and Webb III, T., 1985. Relationships between contemporary pollen and vegetation data from Wisconsin and Michigan, U.S.A. *Ecology*, 66: 721-737.
- Cushing, E.J., 1967. Late-Wisconsin pollen stratigraphy and the glacial sequence in Minnesota. In: E.J. Cushing and H.E. Wright, Jr. (Editors), *Quaternary Paleoecology*. Yale Univ. Press, New Haven, CT, pp.59-88.
- Darwin, C., 1859. *On the Origin of Species by Means of Natural Selection*. Murray, London.
- Davis, M.B., 1976. Pleistocene biogeography of temperate deciduous forests. *Geosci. Man*, 13: 13-26.
- Davis, M.B., 1978. Climatic interpretation of fossil pollen data. In: D. Walker and J.C. Guppy (Editors), *Biology and Quaternary Environments*. Australian Academy of Science, Canberra, pp.35-51.
- Davis, M.B., 1981. Quaternary history and the stability of deciduous forests. In: D.C. West, H.H. Shugart and D.B. Botkin (Editors), *Forest Succession*. Springer, New York, N.Y., pp.132-177.
- Davis, M.B., 1983. Holocene vegetational history of the eastern United States. In: H.E. Wright, Jr. (Editor), *Late Quaternary Environments of the United States*. Vol.2. The Holocene. Univ. Minnesota Press, Minneapolis, Minn., pp.166-181.
- Davis, M.B., Woods, K.D., Webb, S.D. and Futyma, R.P., 1986. Dispersal versus climate: expansion of *Fagus* and *Tsuga* into the upper Great Lakes region. *Vegetatio*, 67: 93-103.
- Delcourt, P.A. and Delcourt, H.R., 1981. Vegetation maps for eastern North America: 40,000 yr B.P. to the present. In: R.C. Romans (Editor), *Geobotany II*. Plenum Press, New York, N.Y., pp.123-166.
- Delcourt, P.A., Delcourt, H.R. and Webb III, T., 1984. Atlas of mapped distributions of dominance and modern pollen percentages for important tree taxa of eastern North America. *Am. Assoc. Stratigr. Palynol. Contrib. Ser.*, 14: 1-130.

- Gaudreau, D.C., 1986. Late-Quaternary vegetational history of the Northeast: paleoecological implications of topographic patterns in pollen distributions. Thesis. Yale Univ., New Haven, Conn.
- Gaudreau, D.C. and Webb III, T., 1985. Late-Quaternary pollen stratigraphy and isochrone maps for the northeastern United States. In: V.M. Bryant Jr. and R.G. Holloway (Editors), *Pollen Records of Late Quaternary North American Sediments*. Am. Assoc. Stratigr. Palynol. 247–280.
- Good, R.D'O., 1931. A theory of plant geography. *New Phytol.*, 30: 149–171.
- Gregory, P.H., 1973. *The Microbiology of the Atmosphere*. Inter-Science, New York, N.Y.
- Huntley, B. and Birks, H.J.B., 1983. *An Atlas of Past and Present Pollen Maps for Europe: 0–13,000 Years Ago*. Cambridge Univ. Press, Cambridge, 667 pp.
- Johnson, W.C. and Atkisson, C.S., 1985. Dispersal of beech nuts by bluejays in fragmented landscapes. *Am. Midl. Nat.*, 113: 319–324.
- Kapp, R.O., 1977. Late Pleistocene and postglacial plant communities of the Great Lakes region. In: R.C. Romans (Editor), *Geobotany*. Plenum Press, New York, N.Y., pp.1–27.
- Karieva, P., 1982. Experimental and mathematical analyses of herbivore movement: quantifying the influence of plant spacing and quality of foraging discrimination. *Ecol. Monogr.*, 52: 261–282.
- Ledbetter, M.T. and Balsam, W.L., 1985. Paleoceanography of the deep western boundary undercurrent on the North American continental margin for the past 25,000 yr. *Geology*, 13: 181–184.
- Levin, S., 1981. The role of theoretical ecology in the description and understanding of populations in heterogeneous environments. *Am. Zool.*, 21: 865–875.
- McAndrews, J.H., 1966. Postglacial history of prairie, savanna and forest in northeastern Minnesota. *Torrey Bot. Club Mem.*, 22: 1–72.
- Okubo, A., 1980. *Diffusion and Ecological Problems: Mathematical Models*. Springer, Berlin, 243 pp.
- Peterson, G.M., 1983. Holocene vegetation and climate in the western U.S.S.R. Thesis. Univ. Wisconsin, Madison, Wisc.
- Prentice, I.C., 1983. Postglacial climatic change: vegetation and the pollen record. *Progr. Phys. Geogr.*, 7: 273–286.
- Prentice, I.C., 1985. Pollen representation, source area and basin size: towards a unified theory of pollen analysis. *Quat. Res.*, 23: 76–86.
- Prentice, I.C., 1986. Vegetation responses to past climatic variation: mechanisms and rates. *Vegetatio*, 67: 131–141.
- Prentice, I.C. and Webb III, T., 1986. Pollen percentages, tree abundances and the Fagerlind effect. *J. Quat. Sci.*, 1 (in press).
- Ritchie, J.C., 1986. Vegetation–climate equilibrium in relation to paleoecology. *Vegetatio*, 67: 65–74.
- Ruddiman, W.F. and McIntyre, A., 1976. Northeast Atlantic paleo-climatic changes over the past 600,000 years. *Geol. Soc. Am. Mem.*, 145: 111–146.
- Skellam, J.G., 1951. Random dispersal in theoretical populations. *Biometrika*, 38: 196–218.
- Solomon, A.M. and Harrington Jr., J.B., 1979. Palynology models. In: R.L. Edmonds (Editor), *Aerobiology: The Ecological Systems Approach*. Dowden, Hutchinson and Ross, Stroudsburg, Pa., pp.338–361.
- Stead, S.E., 1983. Smooth multistage multivariate approximation. Thesis. Dep. Applied Mathematics, Brown Univ., Providence, RI.
- Stead, S. and Webb III, T., in prep. Fitting three dimensional surfaces to pollen percentages plotted in both space and time.
- Tsukada, M., 1982. Late Quaternary development of the *Fagus* forest in the Japanese archipelago. *Jap. J. Ecol.*, 32: 113–118.
- Webb III, T., 1986. Is vegetation in equilibrium with climate? How to interpret late-Quaternary pollen data. *Vegetatio*, 67: 75–91.
- Webb III, T., Cushing, E.J. and Wright Jr., H.E., 1983a. Holocene changes in the vegetation of the Midwest. In: H.E. Wright, Jr. (Editor), *Late-Quaternary Environments of the United States, Vol. 2. The Holocene*, Univ. Minnesota Press, Minneapolis, Minn., pp.142–165.
- Webb III, T., Richard, P. and Mott, R.J., 1983b. A mapped history of Holocene vegetation in southern Quebec. *Syllogeus*, 49: 273–336.
- Wright Jr., H.E. and Watts, W.A., 1969. Glacial and vegetational history of northeastern Minnesota. *Minn. Geol. Surv.*, SP-11: 1–59.

Note added in proof

Tsukada (1982) used a logistic growth model to estimate a value for α of 1.7/1000 yr for *Fagus* pollen percentages from Japanese sites. This value is $10 \times$ higher than the α that we calculated, but Tsukada used unsmoothed data from individual sites and his model had no terms for spatial

variation. For his calculation, he assumed that unchanging optimal environment conditions already existed at the sites. Figure 8 shows that an α of 1.7/1000 yr would lead to values of D and V that were $10 \times$ larger than those that we calculated. Use of unsmoothed data in future work could produce different estimates for α , D , and V than those that we calculated and require use of nonconstant coefficient models.